Finite-Prime Toeplitz Dynamics, KMS States, and Euler Factors

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Abstract

The Bost–Connes system is a paradigmatic example of an arithmetic C*-dynamical system whose equilibrium structure is governed by the Riemann zeta function. In this paper we isolate and analyze an elementary finite-prime model that retains the Euler-product mechanism while avoiding the additional symmetry and adelic structure of the full Bost–Connes system.

Let S be a finite set of rational primes and let

$$\Lambda_S = \left\{ \prod_{p \in S} p^{k_p} : k_p \in \mathbb{N} \right\}.$$

On $\ell^2(\Lambda_S)$ we consider the multiplicative shifts $V_m e_n = e_{mn}$, the Toeplitz algebra $\mathcal{T}_S = C^*(1, V_p : p \in S)$, and the dynamics $\alpha_t(V_m) = m^{it} V_m$. For every real $\beta > 0$ we prove that $(\mathcal{T}_S, \alpha)$ has a unique KMS_β state, given on the standard monomials by

$$\varphi_{\beta,S}(V_m V_n^*) = \delta_{m,n}\, m^{-\beta}.$$

Its Gibbs partition function is

$$Z_S^{\mathrm{fin}}(\beta) = \sum_{n \in \Lambda_S} n^{-\beta} = \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

We then recover the archimedean factor

$$Z_\infty(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right)$$

from the Mellin transform of the Gaussian. The resulting product

$$Z_S(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right) \prod_{p \in S} (1 - p^{-\beta})^{-1}$$

is a finite-prime truncation of the completed zeta factor. The finite-prime factor is a genuine trace-class Gibbs partition function; the archimedean factor is a Mellin normalization, and we keep these two constructions conceptually distinct.

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1 Introduction

The interaction between operator algebras, equilibrium statistical mechanics, and arithmetic became particularly striking in the work of Bost and Connes [1]. Their C*-dynamical system has the Riemann zeta function as its partition function and exhibits a nontrivial KMS phase structure tied to arithmetic symmetry. The present paper studies a much simpler object. We retain only finitely many prime modes and work directly with the Toeplitz shifts generated by multiplication by those primes.

The basic observation is elementary but structurally suggestive. For a prime p and real $\beta > 0$,

$$\sum_{k=0}^{\infty} e^{-\beta k \log p} = \sum_{k=0}^{\infty} p^{-\beta k} = (1 - p^{-\beta})^{-1}.$$

Thus the local Euler factor is the Gibbs partition function of a single occupation-number mode with energy spacing $\log p$. For a finite set of primes the unique-factorization theorem makes the modes independent: if

$$n = \prod_{p \in S} p^{k_p},$$

then

$$\log n = \sum_{p \in S} k_p \log p.$$

The finite Euler product is therefore expected to arise from a diagonal Hamiltonian with energy $\log n$.

Our aim is to make that statement precise at the level of a C*-dynamical system. Let S be a finite set of rational primes and define

$$\Lambda_S := \left\{ \prod_{p \in S} p^{k_p} : k_p \in \mathbb{N} \right\}, \tag{1}$$

where $\mathbb{N} = \{0, 1, 2, \ldots\}$. Thus $1 \in \Lambda_S$, and if $S = \varnothing$ then $\Lambda_S = \{1\}$. We represent this semigroup by multiplicative shifts on $\ell^2(\Lambda_S)$, form the Toeplitz algebra generated by the prime shifts, and equip it with the dynamics determined by the arithmetic energy $\log n$.

The main result is that for every real inverse temperature $\beta > 0$ this finite-prime system has a unique KMS state, with

$$\varphi_{\beta,S}(V_m V_n^*) = \begin{cases} m^{-\beta}, & m = n, \\ 0, & m \neq n, \end{cases}$$

and partition function

$$Z_S^{\mathrm{fin}}(\beta) = \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

For fixed finite S, this Euler product is holomorphic in the larger half-plane $\Re(s) > 0$. The familiar restriction $\Re(s) > 1$ appears only when the finite set S is allowed to exhaust all rational primes, in which case the Euler products converge to $\zeta(s)$.

It is useful to keep this finite-level distinction in view. The full Bost–Connes algebra carries additional arithmetic symmetry and a genuinely nontrivial phase structure, while infinite-prime Toeplitz constructions that recover the full Euler product must confront convergence of the product over all primes. At fixed finite S, by contrast, there is no prime tail: for each $p \in S$, the local factor $(1 - p^{-s})^{-1}$ is holomorphic on $\Re(s) > 0$ because $|p^{-s}| = p^{-\Re(s)} < 1$, and a finite product of

such factors remains holomorphic there. Thus the boundary $\Re(s) = 1$ is a global feature of the infinite-prime limit, not an obstruction already present at finite level.

We also include the real place. The Mellin transform of the Gaussian gives

$$2\int_0^\infty x^{s-1}e^{-\pi x^2}\,dx = \pi^{-s/2}\Gamma\left(\frac{s}{2}\right), \qquad \Re(s) > 0,$$

the standard archimedean factor familiar from the adelic treatment of zeta and L-functions; see, for example, Tate's thesis [6]. Combining this Mellin normalization with the finite-prime Gibbs factor gives a finite truncation of the completed zeta factor.

The model studied here is inspired by the arithmetic quantum statistical mechanics of Bost–Connes, but it is not asserted to be a finite-stage subsystem, quotient, or crossed-product presentation of the full Bost–Connes algebra. Its purpose is narrower: to isolate the finite Euler factors in a completely explicit Toeplitz–KMS system for which existence, uniqueness, and the partition function can all be proved directly.

2 C*-dynamical systems and the KMS condition

We briefly recall the operator-algebraic language used below. Standard references for C*-dynamical systems and KMS states include Bratteli–Robinson [2]; for KMS states on Toeplitz and Nica–Toeplitz algebras, see also [4, 5].

A C*-dynamical system is a pair (A, α), where A is a C*-algebra and

$$\alpha : \mathbb{R} \longrightarrow \mathrm{Aut}(A)$$

is a point-norm continuous one-parameter group:

$$\alpha_{t+u} = \alpha_t \circ \alpha_u, \qquad \|\alpha_t(a) - a\| \longrightarrow 0 \quad (t \to 0)$$

for every $a \in A$.

An element $a \in A$ is entire analytic for α if the map $t \mapsto \alpha_t(a)$ extends to an entire A-valued function $z \mapsto \alpha_z(a)$ on $\mathbb{C}$. Entire analytic elements form a dense $*$-subalgebra. For a real inverse temperature $\beta > 0$, a state φ is a KMS_β state if

$$\varphi(ab) = \varphi(b\,\alpha_{i\beta}(a)) \tag{2}$$

on a dense $*$-algebra of entire analytic elements. This analytic-element form of the KMS condition is equivalent to the standard strip formulation under the usual continuity hypotheses [2].

When the dynamics is implemented on a Hilbert space by a self-adjoint Hamiltonian H,

$$\alpha_t(a) = e^{itH}ae^{-itH},$$

and $e^{-\beta H}$ is trace class, the normalized Gibbs functional

$$\varphi_\beta(a) = \frac{\mathrm{Tr}(e^{-\beta H}a)}{\mathrm{Tr}(e^{-\beta H})}$$

is the canonical KMS state. The finite-prime model below is an infinite-dimensional but completely explicit instance of this mechanism.

3 The finite-prime Toeplitz system

Let

$$\mathcal{H}_S := \ell^2(\Lambda_S),$$

with orthonormal basis $\{e_n : n \in \Lambda_S\}$. For $m \in \Lambda_S$, define an isometry

$$V_m e_n := e_{mn}.$$

Then

$$V_m V_n = V_{mn}, \qquad V_1 = 1, \qquad V_m^* V_m = 1.$$

Unique factorization gives a canonical identification

$$\Lambda_S \cong \prod_{p \in S} \mathbb{N}, \qquad (k_p)_{p \in S} \longmapsto \prod_{p \in S} p^{k_p}.$$

Consequently, if $m = \prod_{p \in S} p^{k_p}$, then

$$V_m = \prod_{p \in S} V_p^{k_p}.$$

Thus the prime shifts already generate all semigroup shifts. Let

$$\mathcal{T}_S := C^*(1, V_p : p \in S) \subset B(\mathcal{H}_S).$$

For a single prime this is the classical Toeplitz algebra generated by a proper isometry, in the sense of Coburn [3]. For general finite S, the identification $\Lambda_S \cong \mathbb{N}^{|S|}$ exhibits Λ_S as the free abelian monoid on the primes in S, with the isometries V_p as its coordinate unilateral shifts. Thus $\mathcal{T}_S$ is a concrete multivariable Toeplitz algebra of the type appearing in the modern Nica–Toeplitz/KMS literature [5]. With this convention, if $S = \varnothing$, then $\mathcal{H}_S = \mathbb{C}$ and $\mathcal{T}_S = \mathbb{C}$. The monomial calculus in this representation is especially simple.

Lemma 1 (Monomial normal form)**.** *Let $n, r \in \Lambda_S$, and put $d = \gcd(n, r)$. Then $d, n/d, r/d \in \Lambda_S$ and*

$$V_n^* V_r = V_{r/d} V_{n/d}^*. \tag{3}$$

Consequently, for $m, n, r, s \in \Lambda_S$,

$$(V_m V_n^*)(V_r V_s^*) = V_{mr/d} V_{sn/d}^*, \qquad d = \gcd(n, r). \tag{4}$$

Proof. Under the canonical identification

$$\Lambda_S \cong \prod_{p \in S} \mathbb{N},$$

write

$$n = \prod_{p \in S} p^{k_p}, \qquad r = \prod_{p \in S} p^{\ell_p}.$$

Then

$$d = \gcd(n, r) = \prod_{p \in S} p^{\min(k_p, \ell_p)}.$$

In particular $d \in \Lambda_S$, and

$$\frac{n}{d} = \prod_{p \in S} p^{k_p - \min(k_p, \ell_p)}, \qquad \frac{r}{d} = \prod_{p \in S} p^{\ell_p - \min(k_p, \ell_p)}$$

also belong to Λ_S.

Put

$$n_0 = \frac{n}{d}, \qquad r_0 = \frac{r}{d}.$$

For every $p \in S$, at least one of the two residual exponents

$$k_p - \min(k_p, \ell_p), \qquad \ell_p - \min(k_p, \ell_p)$$

is zero. Thus the residual exponent vectors have disjoint support, and hence

$$\gcd(n_0, r_0) = 1.$$

For $k \in \Lambda_S$,

$$V_n^* e_k = \begin{cases} e_{k/n}, & n \mid k, \\ 0, & n \nmid k. \end{cases}$$

Since $n = dn_0$ and $r = dr_0$, with $\gcd(n_0, r_0) = 1$,

$$n \mid rk \quad \Longleftrightarrow \quad n_0 \mid r_0 k \quad \Longleftrightarrow \quad n_0 \mid k.$$

Thus $V_n^* V_r$ and $V_{r_0} V_{n_0}^*$ have the same action on every basis vector e_k, proving (3). The product formula follows from

$$V_m (V_n^* V_r) V_s^* = V_m V_{r/d} V_{n/d}^* V_s^* = V_{mr/d} V_{sn/d}^*.$$

□

It is useful to isolate the two algebraic mechanisms behind the preceding normal form. If $d, a, b \in \Lambda_S$, then cancellation of a common middle factor gives

$$V_{da}^* V_{db} = V_a^* V_b.$$

If in addition $\gcd(a, b) = 1$, then the residual shifts doubly commute:

$$V_a^* V_b = V_b V_a^*.$$

Thus, writing $n = da$ and $r = db$ with $d = \gcd(n, r)$, the normal-form identity factors conceptually as

$$V_n^* V_r = V_{da}^* V_{db} = V_a^* V_b = V_b V_a^*,$$

which is exactly

$$V_n^* V_r = V_{r/d} V_{n/d}^*.$$

Define the algebraic monomial core

$$\mathcal{T}_S^{\mathrm{alg}} := \mathrm{span}_{\mathbb{C}} \{ V_m V_n^* : m, n \in \Lambda_S \}. \tag{5}$$

Lemma 2 (Algebraic monomial core and unique expansion)**.** *The algebraic $*$-algebra generated by the prime shifts V_p, $p \in S$, is exactly $\mathcal{T}_S^{\mathrm{alg}}$. The family*

$$\{ V_m V_n^* : m, n \in \Lambda_S \}$$

is linearly independent. Consequently every $A \in \mathcal{T}_S^{\mathrm{alg}}$ has a unique finite expansion

$$A = \sum_{(m,n) \in F} c_{m,n} V_m V_n^*,$$

where $F \subset \Lambda_S \times \Lambda_S$ is finite and $c_{m,n} \in \mathbb{C}$. Moreover $\mathcal{T}_S^{\mathrm{alg}}$ is norm-dense in $\mathcal{T}_S$.

Proof. Equation (4) shows that the span in (5) is closed under multiplication, and taking adjoints interchanges m and n. It contains $1 = V_1 V_1^*$ and every prime generator $V_p = V_p V_1^*$. Conversely, every monomial lies in the algebra generated by the prime shifts and their adjoints. Hence $\mathcal{T}_S^{\mathrm{alg}}$ is precisely that unital $*$-algebra, and its norm closure is $\mathcal{T}_S$. Existence of a finite expansion is the defining property of the algebraic span.

For uniqueness it suffices to prove linear independence. Suppose that a finite relation

$$\sum_{(r,s)\in F} c_{r,s} V_r V_s^* = 0$$

holds. We prove by strong induction on the positive integer n that every coefficient $c_{m,n}$ occurring in the relation is zero. Fix m, n and assume that all coefficients with second index $s < n$ have already been shown to vanish. Apply the displayed operator relation to e_n and take the e_m-coordinate. A term $V_r V_s^* e_n$ can be nonzero only if $s \mid n$. If $s \mid n$ and $s < n$, its coefficient vanishes by the induction hypothesis. If $s = n$, then

$$V_r V_n^* e_n = e_r,$$

so its e_m-coordinate is nonzero only for $r = m$, in which case it is one. No divisor $s \mid n$ can satisfy $s > n$. Thus the chosen matrix coefficient of the relation is simply $c_{m,n}$, and hence $c_{m,n} = 0$. Strong induction proves that all coefficients vanish. Therefore the monomials are linearly independent and the finite expansion is unique. □

Rather than beginning with an unbounded Hamiltonian, define the dynamics directly by the diagonal unitaries

$$U_t e_n := n^{it} e_n = e^{it \log n} e_n, \qquad t \in \mathbb{R}. \tag{6}$$

Since $|n^{it}| = 1$, each U_t is unitary, and

$$U_0 = 1, \qquad U_{t+u} = U_t U_u, \qquad U_t^* = U_{-t}.$$

Define

$$\alpha_t(a) := U_t a U_{-t}, \qquad a \in B(\mathcal{H}_S).$$

A direct calculation on basis vectors gives

$$\alpha_t(V_m) = m^{it} V_m. \tag{7}$$

Proposition 1. *For each $t \in \mathbb{R}$, the automorphism α_t preserves $\mathcal{T}_S$, and its restriction*

$$\alpha_t : \mathcal{T}_S \xrightarrow{\sim} \mathcal{T}_S$$

is a $*$-automorphism. The family $(\alpha_t)_{t\in\mathbb{R}}$ satisfies

$$\alpha_{t+u} = \alpha_t \circ \alpha_u, \qquad \alpha_0 = \mathrm{id},$$

and is point-norm continuous. Hence $(\mathcal{T}_S, \alpha)$ is a C^*-dynamical system.

Proof. Equation (7) shows that every prime generator V_p is sent to the scalar multiple $p^{it} V_p$. Since α_t is a $*$-automorphism of $B(\mathcal{H}_S)$, it follows that the algebraic $*$-algebra generated by the V_p's is invariant. Conjugation by a unitary is isometric, hence norm-continuous, so α_t preserves its norm closure $\mathcal{T}_S$. Applying the same argument to α_{-t} gives equality rather than merely inclusion.

The group law follows from $U_{t+u} = U_t U_u$. For point-norm continuity, first note that

$$\alpha_t(V_m V_n^*) = m^{it} n^{-it} V_m V_n^* = \left(\frac{m}{n}\right)^{it} V_m V_n^*,$$

so $t \mapsto \alpha_t(a)$ is norm-continuous for every element a in the algebraic span of the monomials. Given arbitrary $a \in \mathcal{T}_S$ and $\varepsilon > 0$, choose such a finite linear combination b with $\|a - b\| < \varepsilon/3$. Since every α_t is isometric,

$$\|\alpha_t(a) - \alpha_s(a)\| \le 2\|a - b\| + \|\alpha_t(b) - \alpha_s(b)\|.$$

For t sufficiently close to s, continuity on the algebraic span gives $\|\alpha_t(b) - \alpha_s(b)\| < \varepsilon/3$. Hence the displayed quantity is strictly less than ε, proving point-norm continuity. □

Proposition 2 (Entire analytic core)**.** *Every element of $\mathcal{T}_S^{\mathrm{alg}}$ is entire analytic for α. More precisely, for $m, n \in \Lambda_S$,*

$$\alpha_z(V_m V_n^*) := e^{iz(\log m - \log n)} V_m V_n^*, \qquad z \in \mathbb{C}, \tag{8}$$

defines the entire extension of $t \mapsto \alpha_t(V_m V_n^*)$. If

$$A = \sum_{(m,n)\in F} c_{m,n} V_m V_n^*$$

is the unique finite expansion from Lemma 2, then

$$\alpha_z(A) = \sum_{(m,n)\in F} c_{m,n} e^{iz(\log m - \log n)} V_m V_n^*, \qquad z \in \mathbb{C},$$

is therefore an intrinsic, representation-independent entire continuation. Consequently, $\mathcal{T}_S^{\mathrm{alg}}$ is a dense unital $*$-subalgebra of entire analytic elements.

Proof. For real t, equation (7) and its adjoint give

$$\alpha_t(V_m V_n^*) = e^{it(\log m - \log n)} V_m V_n^*.$$

The scalar function $z \mapsto e^{iz(\log m - \log n)}$ is entire, so (8) is an entire norm-valued extension. By Lemma 2, every element of the algebraic core has a unique finite monomial expansion, so the displayed termwise formula is well-defined independently of any choice of representation. Finite sums of entire norm-valued functions remain entire. Density and the $*$-algebra property were established above. □

The logarithmic Hamiltonian. Define the diagonal operator H_S by

$$(H_S \xi)_n = (\log n)\xi_n, \qquad \mathrm{Dom}(H_S) = \left\{ \xi \in \ell^2(\Lambda_S) : \sum_{n\in\Lambda_S} (\log n)^2 |\xi_n|^2 < \infty \right\}.$$

Every finite linear combination of the basis vectors belongs to this domain, so H_S is densely defined. In particular, $H_S e_n = (\log n) e_n$, and $H_S e_1 = 0$. For $\xi, \eta \in \mathrm{Dom}(H_S)$, the relevant coordinate inner-product series converge absolutely by the Cauchy–Schwarz inequality. Since every $\log n$ is real,

$$\langle H_S \xi, \eta \rangle = \langle \xi, H_S \eta \rangle.$$

Moreover, because $n \ge 1$,

$$\langle \xi, H_S \xi \rangle = \sum_{n\in\Lambda_S} (\log n) |\xi_n|^2 \ge 0.$$

Thus H_S is symmetric and positive on its natural domain.

To prove self-adjointness, let $\eta \in \mathrm{Dom}(H_S^*)$ and write $\zeta = H_S^*\eta$. Testing the adjoint identity on e_n gives

$$\zeta_n = (\log n)\eta_n, \qquad n \in \Lambda_S.$$

Since $\zeta \in \ell^2(\Lambda_S)$, it follows that

$$\sum_{n\in\Lambda_S} (\log n)^2|\eta_n|^2 = \|\zeta\|^2 < \infty.$$

Hence $\eta \in \mathrm{Dom}(H_S)$ and $H_S\eta = \zeta$. Conversely, symmetry gives $H_S \subseteq H_S^*$. Therefore

$$\mathrm{Dom}(H_S^*) = \mathrm{Dom}(H_S), \qquad H_S^* = H_S.$$

In particular, H_S is closed. Here closedness refers to its graph; it does not assert that its domain is closed in the ambient Hilbert norm.

The diagonal spectral calculus now gives $e^{itH_S}e_n = e^{it\log n}e_n = U_t e_n$, so $U_t = e^{itH_S}$. The group (U_t) is strongly continuous: this follows first on finite basis sums from scalar continuity and then on all of $\mathcal{H}_S$ by density and $\|U_t\| = 1$. Thus H_S is its self-adjoint generator. This Hamiltonian description is useful for Gibbs weights and traces, but the bounded unitaries (6) provide the cleaner definition of the dynamics itself. The logarithmic energy assignment is the same arithmetic scaling mechanism that underlies the time evolution in the Bost–Connes system [1], here restricted to the finite semigroup Λ_S.

Proposition 3 (Domain invariance and strong differentiation)**.** *For every $t \in \mathbb{R}$,*

$$U_t\,\mathrm{Dom}(H_S) = \mathrm{Dom}(H_S), \qquad H_S U_t\xi = U_t H_S\xi \quad (\xi \in \mathrm{Dom}(H_S)).$$

For each $\xi \in \mathrm{Dom}(H_S)$, the orbit $t \mapsto U_t\xi$ is continuous in the graph norm $\|\xi\|_{H_S} = (\|\xi\|^2 + \|H_S\xi\|^2)^{1/2}$ and continuously differentiable in the Hilbert-space norm, with

$$\frac{d}{dt}U_t\xi = iU_tH_S\xi = iH_SU_t\xi.$$

In particular, the strong infinitesimal generator acts as iH_S on $\mathrm{Dom}(H_S)$, consistently with $U_t = e^{itH_S}$.

Proof. Since $(U_t\xi)_n = e^{it\log n}\xi_n$ and $|e^{it\log n}| = 1$, the weighted square sum defining $\mathrm{Dom}(H_S)$ is unchanged by U_t. Applying this also to U_{-t} gives equality of domains. Multiplication by $\log n$ commutes coordinatewise with the phase, which proves $H_SU_t\xi = U_tH_S\xi$. Strong continuity applied to both ξ and $H_S\xi$ now gives graph-norm continuity.

For $h \neq 0$, the elementary estimate $|e^{ia} - 1| \le |a|$, $a \in \mathbb{R}$, gives

$$\left|\frac{e^{ih\log n} - 1}{h} - i\log n\right| |\xi_n| \le 2(\log n)|\xi_n|.$$

The expression on the left tends to zero for each n. Its square is dominated by $4(\log n)^2|\xi_n|^2$, a summable sequence. Dominated convergence for series therefore gives

$$\lim_{h\to 0}\left\|\frac{U_h\xi - \xi}{h} - iH_S\xi\right\| = 0.$$

The group law yields the derivative at arbitrary t by applying the bounded operator U_t to this limit. Commutation gives the second expression for the derivative, and strong continuity of $t \mapsto U_tH_S\xi$ shows that the derivative is continuous. □

4 The finite Euler product as a trace partition function

Lemma 3. *For every real $\beta > 0$, the operator $e^{-\beta H_S}$ is trace class and*

$$Z_S^{\mathrm{fin}}(\beta) := \mathrm{Tr}(e^{-\beta H_S}) = \sum_{n \in \Lambda_S} n^{-\beta} = \prod_{p \in S} (1 - p^{-\beta})^{-1}. \tag{9}$$

Moreover, the finite Euler product

$$s \longmapsto \prod_{p \in S} (1 - p^{-s})^{-1}$$

is holomorphic on the half-plane $\Re(s) > 0$.

Proof. Since H_S is positive and self-adjoint, $e^{-\beta H_S}$ is a bounded positive diagonal operator, with

$$e^{-\beta H_S} e_n = n^{-\beta} e_n.$$

We first establish summability of these diagonal entries. Unique factorization identifies Λ_S with $\prod_{p \in S} \mathbb{N}$. Since all terms are nonnegative, Tonelli's theorem for series gives

$$\sum_{n \in \Lambda_S} n^{-\beta} = \sum_{(k_p) \in \prod_{p \in S} \mathbb{N}} \prod_{p \in S} p^{-\beta k_p} = \prod_{p \in S} \sum_{k=0}^{\infty} p^{-\beta k} = \prod_{p \in S} (1 - p^{-\beta})^{-1} < \infty.$$

Each geometric series converges because $0 < p^{-\beta} < 1$. A positive diagonal operator is trace class precisely when its diagonal entries are summable. Applying this criterion now proves that $e^{-\beta H_S}$ is trace class and that

$$\mathrm{Tr}(e^{-\beta H_S}) = \sum_{n \in \Lambda_S} n^{-\beta} = \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

Equivalently, each rank-one term $n^{-\beta} |e_n\rangle\langle e_n|$ has trace norm $n^{-\beta}$, so the diagonal expansion converges absolutely in trace norm to $e^{-\beta H_S}$.

For a complex variable s with $\Re(s) > 0$, each local geometric series converges to the holomorphic factor $(1 - p^{-s})^{-1}$; since S is finite, their product is holomorphic throughout that half-plane. For the real inverse-temperature parameter, if $S \neq \varnothing$, the endpoint $\beta = 0$ is sharp: then $n^{-\beta} = 1$ for every $n \in \Lambda_S$, and the infinite sum diverges. If $S = \varnothing$, then $\Lambda_S = \{1\}$ and the partition function is identically 1. This proves the claim. □

5 Existence and uniqueness of the finite-prime KMS state

For $\beta > 0$, it is convenient to write the positive diagonal Gibbs operator as

$$D_{\beta,S} := e^{-\beta H_S} = \sum_{n \in \Lambda_S} n^{-\beta} \, |e_n\rangle\langle e_n|, \tag{10}$$

where the series converges in trace norm (and hence in operator norm). Its trace is exactly the finite partition function,

$$\mathrm{Tr}(D_{\beta,S}) = Z_S^{\mathrm{fin}}(\beta).$$

Thus the normalized coefficients

$$\mu_{\beta,S}(n) := \frac{n^{-\beta}}{Z_S^{\mathrm{fin}}(\beta)}, \qquad n \in \Lambda_S, \tag{11}$$

form a probability distribution on Λ_S:

$$\mu_{\beta,S}(n) \geq 0, \qquad \sum_{n \in \Lambda_S} \mu_{\beta,S}(n) = 1.$$

The factorization of this probability law mirrors the factorization of the partition function.

Proposition 4 (Independent geometric prime exponents)**.** *Let $\beta > 0$, and write*

$$n = \prod_{p \in S} p^{k_p}.$$

Then

$$\mu_{\beta,S}(n) = \prod_{p \in S} (1 - p^{-\beta}) p^{-\beta k_p}. \tag{12}$$

Equivalently, if K_p, $p \in S$, are independent $\mathbb{N}$-valued random variables with

$$\Pr(K_p = k) = (1 - p^{-\beta}) p^{-\beta k}, \qquad k \in \mathbb{N},$$

then the random arithmetic state

$$N_S := \prod_{p \in S} p^{K_p}$$

has law $\mu_{\beta,S}$ on Λ_S.

Proof. By Lemma 3,

$$\frac{1}{Z_S^{\mathrm{fin}}(\beta)} = \prod_{p \in S} (1 - p^{-\beta}).$$

Since $n^{-\beta} = \prod_{p \in S} p^{-\beta k_p}$, substitution into (11) gives (12). Each factor is the geometric probability mass function on $\mathbb{N} = \{0, 1, 2, \ldots\}$ with ratio $p^{-\beta}$, and unique factorization identifies the product law on the exponent vector $(K_p)_{p \in S}$ with the stated law on Λ_S. For $S = \varnothing$, the empty product gives the point mass at 1. □

Define

$$\varphi_{\beta,S}(a) := \frac{\mathrm{Tr}(D_{\beta,S} a)}{Z_S^{\mathrm{fin}}(\beta)}, \qquad a \in \mathcal{T}_S. \tag{13}$$

This is a state because $D_{\beta,S}$ is positive trace class and its trace is $Z_S^{\mathrm{fin}}(\beta)$. Equivalently, the Gibbs state is the weighted basis-diagonal expectation

$$\varphi_{\beta,S}(a) = \sum_{n \in \Lambda_S} \mu_{\beta,S}(n) \, \langle e_n, a e_n \rangle. \tag{14}$$

In particular, one may regard $\varphi_{\beta,S}$ as averaging the diagonal matrix coefficient of a against the Gibbs probability distribution on the arithmetic states $n \in \Lambda_S$.

Theorem 1. *For every real $\beta > 0$, the state $\varphi_{\beta,S}$ is the unique* KMS_β *state on the finite-prime Toeplitz system $(\mathcal{T}_S, \alpha)$. On the dense $*$-subalgebra spanned by $V_m V_n^*$, it is given by*

$$\varphi_{\beta,S}(V_m V_n^*) = \begin{cases} m^{-\beta}, & m = n, \\ 0, & m \neq n. \end{cases} \tag{15}$$

Proof. First we verify that $\varphi_{\beta,S}$ has the values claimed in (15). For $k \in \Lambda_S$,

$$\langle e_k, V_m V_n^* e_k \rangle$$

can be nonzero only if n divides k inside Λ_S and $mk/n = k$. The latter equality forces $m = n$. Therefore, if $m \neq n$,

$$\mathrm{Tr}(e^{-\beta H_S} V_m V_n^*) = 0.$$

If $m = n$, then $V_m V_m^*$ is the projection onto the closed span of $\{e_{mr} : r \in \Lambda_S\}$. Hence

$$\mathrm{Tr}(e^{-\beta H_S} V_m V_m^*) = \sum_{r \in \Lambda_S} (mr)^{-\beta} = m^{-\beta} Z_S^{\mathrm{fin}}(\beta).$$

Dividing by $Z_S^{\mathrm{fin}}(\beta)$ gives (15).

Two elementary consequences of this formula are useful to keep in view. Since $V_n^* V_m$ has Gibbs expectation 1 when $m = n$ and 0 otherwise,

$$\varphi_{\beta,S}(V_n^* V_m) = \delta_{m,n}.$$

Consequently,

$$\varphi_{\beta,S}(V_m V_n^*) = m^{-\beta} \varphi_{\beta,S}(V_n^* V_m).$$

Thus the KMS boundary relation is already visible on a single shift and adjoint shift. The argument below verifies the full KMS identity on the dense algebra of analytic monomials.

The Gibbs state (13) satisfies the KMS_β condition. We verify this directly on the dense entire analytic core $\mathcal{T}_S^{\mathrm{alg}}$ from Proposition 2. Let

$$A = V_m V_n^*.$$

Then

$$\alpha_{i\beta}(A) = e^{-\beta(\log m - \log n)} A = (m/n)^{-\beta} A.$$

Equivalently, the boundary scalar is exactly the ratio of the two Gibbs weights:

$$e^{-\beta(\log m - \log n)} = \frac{m^{-\beta}}{n^{-\beta}}.$$

Thus the KMS boundary factor is literally the imaginary-time value of the entire monomial evolution in (8). Write $D_{\beta,S} = e^{-\beta H_S}$. The commutation relation

$$D_{\beta,S} A = \alpha_{i\beta}(A) D_{\beta,S} \tag{16}$$

is a bounded-operator identity: it may be checked directly on the basis vectors e_k. Indeed, whenever $V_m V_n^* e_k \neq 0$,

$$D_{\beta,S} V_m V_n^* e_k = (mk/n)^{-\beta} V_m V_n^* e_k = (m/n)^{-\beta} V_m V_n^* D_{\beta,S} e_k,$$

and both sides vanish otherwise. Thus no use of the generally unbounded operator $e^{\beta H_S}$ is required.

For analytic monomials A, B, the operator $D_{\beta,S} A$ is trace class, so cyclicity of the trace for a trace-class operator times a bounded operator gives

$$\mathrm{Tr}(D_{\beta,S} AB) = \mathrm{Tr}(B D_{\beta,S} A) = \mathrm{Tr}\big(B \alpha_{i\beta}(A) D_{\beta,S}\big) = \mathrm{Tr}\big(D_{\beta,S} B \alpha_{i\beta}(A)\big).$$

After division by $Z_S^{\mathrm{fin}}(\beta)$, this is precisely

$$\varphi_{\beta,S}(AB) = \varphi_{\beta,S}\big(B \alpha_{i\beta}(A)\big)$$

for analytic monomials A and B.

Now let $A, B \in \mathcal{T}_S^{\mathrm{alg}}$. Using the unique finite expansions from Lemma 2, write

$$A = \sum_{(m,n)\in F} a_{m,n} V_m V_n^*, \qquad B = \sum_{(r,s)\in G} b_{r,s} V_r V_s^*,$$

with $F, G \subset \Lambda_S \times \Lambda_S$ finite. Applying the monomial identity term by term and using bilinearity gives

$$\begin{aligned}
\varphi_{\beta,S}(AB) &= \sum_{(m,n)\in F} \sum_{(r,s)\in G} a_{m,n} b_{r,s}\, \varphi_{\beta,S}\big((V_m V_n^*)(V_r V_s^*)\big) \\
&= \sum_{(m,n)\in F} \sum_{(r,s)\in G} a_{m,n} b_{r,s}\, \varphi_{\beta,S}\big((V_r V_s^*)\alpha_{i\beta}(V_m V_n^*)\big) \\
&= \varphi_{\beta,S}\big(B\alpha_{i\beta}(A)\big).
\end{aligned}$$

Thus the KMS identity holds on the dense $*$-subalgebra $\mathcal{T}_S^{\mathrm{alg}}$ of entire analytic elements, which proves that $\varphi_{\beta,S}$ is a KMS_β state.

It remains to prove uniqueness. Let ψ be any KMS_β state on $(\mathcal{T}_S, \alpha)$. Apply the KMS identity to $a = V_m V_n^*$ and $b = 1$. Since

$$\alpha_{i\beta}(V_m V_n^*) = (m/n)^{-\beta} V_m V_n^*,$$

we obtain

$$\psi(V_m V_n^*) = (m/n)^{-\beta} \psi(V_m V_n^*).$$

If $m \neq n$, then $(m/n)^{-\beta} \neq 1$, so

$$\psi(V_m V_n^*) = 0.$$

If $m = n$, use the KMS condition with $a = V_m$ and $b = V_m^*$. Since

$$\alpha_{i\beta}(V_m) = m^{-\beta} V_m,$$

we obtain

$$\psi(V_m V_m^*) = \psi(V_m^* \alpha_{i\beta}(V_m)) = m^{-\beta} \psi(V_m^* V_m) = m^{-\beta} \psi(1) = m^{-\beta}.$$

Thus ψ agrees with $\varphi_{\beta,S}$ on every standard monomial $V_m V_n^*$. By linearity it therefore agrees with $\varphi_{\beta,S}$ on every finite monomial expansion. By Lemma 2, these finite expansions give the intrinsic monomial coordinates on $\mathcal{T}_S^{\mathrm{alg}}$, so the two functionals agree on the whole algebraic core. Both functionals are bounded, hence norm-continuous, and $\mathcal{T}_S^{\mathrm{alg}}$ is norm-dense in $\mathcal{T}_S$. Therefore $\psi = \varphi_{\beta,S}$ on all of $\mathcal{T}_S$, proving uniqueness. □

Remark 1 (Rigidity does not require positivity)**.** *The uniqueness argument uses only complex linearity, boundedness, the normalization $\psi(1) = 1$, and the KMS boundary identity on the dense algebraic core $\mathcal{T}_S^{\mathrm{alg}}$. Positivity is needed in order to call ψ a state, but it plays no role in the rigidity mechanism itself. Thus any normalized bounded linear functional on $\mathcal{T}_S$ satisfying the same core KMS identity is forced, by the monomial computation, finite expansion, and density, to equal $\varphi_{\beta,S}$.*

6 The archimedean Gamma factor

The finite-prime KMS theorem accounts for the finite Euler product. The factor

$$\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)$$

is the standard archimedean Mellin factor. Its appearance from the Gaussian is classical and is central to the adelic treatment of zeta and L-functions; see Tate [6].

Let

$$d\mu_\infty(x) := 2e^{-\pi x^2}\frac{dx}{x}, \qquad x \in (0,\infty).$$

For real $\beta > 0$, define

$$Z_\infty(\beta) := \int_0^\infty x^\beta \, d\mu_\infty(x) = 2\int_0^\infty x^{\beta-1} e^{-\pi x^2}\, dx. \tag{17}$$

Lemma 4. *For $\Re(s) > 0$,*

$$2\int_0^\infty x^{s-1} e^{-\pi x^2}\, dx = \pi^{-s/2}\Gamma\left(\frac{s}{2}\right). \tag{18}$$

In particular,

$$Z_\infty(\beta) = \pi^{-\beta/2}\Gamma\left(\frac{\beta}{2}\right)$$

for every real $\beta > 0$.

Proof. Using the substitution $u = \pi x^2$, so that

$$x = (u/\pi)^{1/2}, \qquad \frac{dx}{x} = \frac{1}{2}\frac{du}{u},$$

we compute

$$2\int_0^\infty x^s e^{-\pi x^2}\frac{dx}{x} = 2\int_0^\infty \left(\frac{u}{\pi}\right)^{s/2} e^{-u}\frac{1}{2}\frac{du}{u} = \pi^{-s/2}\int_0^\infty u^{s/2-1}e^{-u}\, du.$$

The last integral is $\Gamma(s/2)$, valid for $\Re(s) > 0$. This proves the formula. □

The corresponding normalized archimedean probability measure is

$$d\nu_{\beta,\infty}(x) := \frac{x^\beta}{Z_\infty(\beta)}\, d\mu_\infty(x) = \frac{2x^{\beta-1}e^{-\pi x^2}}{\pi^{-\beta/2}\Gamma(\beta/2)}\, dx. \tag{19}$$

This is a genuine Radon probability measure on $(0,\infty)$ for every real $\beta > 0$.

7 The combined finite-level state and partition function

Define the product algebra using the minimal (spatial) tensor product,

$$\mathcal{A}_S := \mathcal{T}_S \otimes_{\min} C_0((0,\infty)),$$

with dynamics

$$\sigma_t := \alpha_t \otimes \mathrm{id}\,.$$

The algebra $C_0((0,\infty))$, and hence $\mathcal{A}_S$, is non-unital but σ-unital. We use the standard non-unital formulation of the KMS condition: the KMS identity is checked on a dense algebra of entire analytic elements and then extended to the C*-algebra by continuity, equivalently by applying the usual approximate-unit argument; see, for example, [2].

For real $\beta > 0$, define a state on elementary tensors by

$$\omega_{\beta,S}(a \otimes f) := \varphi_{\beta,S}(a)\int_0^\infty f(x)\, d\nu_{\beta,\infty}(x), \tag{20}$$

and extend linearly and continuously to $\mathcal{A}_S$.

Proposition 5. *For every real $\beta > 0$, the state $\omega_{\beta,S}$ is a KMS_β state for $(\mathcal{A}_S, \sigma)$. More generally, every KMS_β state on $(\mathcal{A}_S, \sigma)$ is of the form $\varphi_{\beta,S} \otimes \rho$, where ρ is a state on $C_0((0,\infty))$. Consequently, among KMS states whose archimedean marginal is the probability measure $\nu_{\beta,\infty}$, $\omega_{\beta,S}$ is unique. Its normalizing factor is*

$$Z_S(\beta) = Z_S^{\mathrm{fin}}(\beta) Z_\infty(\beta) = \left(\prod_{p \in S} (1 - p^{-\beta})^{-1} \right) \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right). \tag{21}$$

Proof. The state $\varphi_{\beta,S}$ is KMS_β for $(\mathcal{T}_S, \alpha)$ by Theorem 1. The state $f \mapsto \int f \, d\nu_{\beta,\infty}$ is invariant under the trivial dynamics on $C_0((0,\infty))$. Therefore their tensor product is KMS_β for $\alpha \otimes \mathrm{id}$. Explicitly, for analytic elementary tensors,

$$\omega_{\beta,S}((a \otimes f)\sigma_{i\beta}(b \otimes g)) = \varphi_{\beta,S}(a\alpha_{i\beta}(b)) \int fg \, d\nu_{\beta,\infty},$$

while

$$\omega_{\beta,S}((b \otimes g)(a \otimes f)) = \varphi_{\beta,S}(ba) \int gf \, d\nu_{\beta,\infty}.$$

The two quantities are equal by the finite-prime KMS condition and commutativity of the archimedean factor. Density gives the result for $\mathcal{A}_S$.

The formula for $Z_S(\beta)$ follows immediately by multiplying Lemma 3 and Lemma 4.

It remains to justify the uniqueness statement in the non-unital tensor product. Let ω be any KMS_β state on $(\mathcal{A}_S, \sigma)$. For $0 \le f \in C_0((0,\infty))$, define a positive functional on $\mathcal{T}_S$ by

$$\psi_f(a) := \omega(a \otimes f).$$

If $\psi_f(1) > 0$, then the normalized functional $\psi_f/\psi_f(1)$ is a KMS_β state on $(\mathcal{T}_S, \alpha)$. Indeed, applying the KMS identity in $\mathcal{A}_S$ to $a \otimes f^{1/2}$ and $b \otimes f^{1/2}$ gives

$$\psi_f(a\alpha_{i\beta}(b)) = \psi_f(ba)$$

for analytic $a, b \in \mathcal{T}_S$. By Theorem 1, $\psi_f = \psi_f(1)\varphi_{\beta,S}$. If $\psi_f(1) = 0$, positivity gives $\psi_f = 0$, so the same conclusion holds.

Now set

$$\rho(f) := \omega(1 \otimes f), \qquad f \in C_0((0,\infty)).$$

Then ρ is a positive functional. If (u_j) is a positive contractive approximate unit for $C_0((0,\infty))$, then $1 \otimes u_j$ is an approximate unit for $\mathcal{A}_S$, so $\rho(u_j) \to \|\omega\| = 1$. Hence ρ is a state. The preceding paragraph shows

$$\omega(a \otimes f) = \varphi_{\beta,S}(a)\rho(f)$$

for all $a \in \mathcal{T}_S$ and all positive $f \in C_0((0,\infty))$, and therefore for all f by linearity. Thus $\omega = \varphi_{\beta,S} \otimes \rho$. Taking $\rho(f) = \int f \, d\nu_{\beta,\infty}$ gives the particular state (20), and it is the only KMS state with that archimedean marginal. □

Remark 2 (Can the archimedean factor be made operator-algebraic?)**.** *Yes, but not in quite the same trace-class way as the finite-prime Toeplitz factor. Let*

$$\Phi_\infty(f) := 2 \int_0^\infty f(x) e^{-\pi x^2} \frac{dx}{x}$$

be the natural Gaussian Mellin weight on $C_0((0,\infty))$. If M_x denotes multiplication by x, then formally

$$Z_\infty(\beta) = \Phi_\infty(M_x^\beta)$$

for $\beta > 0$. This is a legitimate commutative operator-algebraic normalization by a semifinite weight, but it is not a discrete trace partition function $\mathrm{Tr}(e^{-\beta H})$ on ℓ^2 unless one adds an additional discretization. In particular, nothing in the present construction identifies the product normalization $Z_S^{\mathrm{fin}}(\beta)Z_\infty(\beta)$ with $\mathrm{Tr}(e^{-\beta H})$ for a single trace-class heat operator on a Hilbert-space representation of $\mathcal{A}_S$. For this reason the paper keeps the finite-prime trace-class Gibbs normalization and the archimedean Mellin normalization conceptually separate.

8 Relation to the Bost–Connes system and scope

The full Bost–Connes system contains substantially more structure than the finite-prime model considered here. In particular, its arithmetic symmetry, its low-temperature extremal KMS states, and the spontaneous symmetry breaking described in [1] are not present in the construction above. We have retained only the multiplicative semigroup generated by a finite set of primes and the logarithmic time evolution that turns multiplication into additive energy.

Nevertheless, the finite model isolates one important mechanism of the arithmetic quantum statistical picture. Each prime contributes a local geometric partition factor

$$Z_p(\beta) = (1 - p^{-\beta})^{-1},$$

and unique factorization makes the finite-prime partition function the product of these independent local factors:

$$Z_S^{\mathrm{fin}}(\beta) = \prod_{p \in S} Z_p(\beta).$$

At fixed finite S, this expression is holomorphic for $\Re(s) > 0$. If S increases through the finite subsets of all rational primes, the Euler products converge for $\Re(s) > 1$ to

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}.$$

Thus the familiar half-plane $\Re(s) > 1$ is a global phenomenon: it is not needed at any fixed finite level.

The archimedean factor should also be interpreted carefully. The finite prime contribution is the trace of a trace-class Gibbs operator on $\ell^2(\Lambda_S)$. By contrast,

$$\pi^{-s/2}\Gamma(s/2)$$

arises here from a Gaussian Mellin transform, as in the classical local analysis of zeta functions [6]. Their product is therefore a natural finite-place-plus-archimedean normalization, but the present paper does not identify it with the trace of one heat operator on a single Hilbert-space representation. Keeping this distinction explicit prevents the analogy with statistical mechanics from being stronger than the construction justifies.

9 Conclusion

For a finite set S of rational primes, we have constructed an explicit C*-dynamical system

$$(\mathcal{T}_S, \alpha), \qquad \mathcal{T}_S = C^*(1, V_p : p \in S), \qquad \alpha_t(V_m) = m^{it} V_m,$$

and proved that for every real $\beta > 0$ it has a unique KMS_β state. On the dense monomial algebra,

$$\varphi_{\beta,S}(V_m V_n^*) = \delta_{m,n} m^{-\beta},$$

and its Gibbs partition function is

$$Z_S^{\mathrm{fin}}(\beta) = \sum_{n \in \Lambda_S} n^{-\beta} = \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

The real place contributes the classical Mellin–Gaussian factor

$$Z_\infty(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right),$$

so the combined normalization is

$$Z_S(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right) \prod_{p \in S} (1 - p^{-\beta})^{-1}.$$

The standard monomials $V_m V_n^*$ are linearly independent, so the algebraic core has an intrinsic finite monomial coordinate system; in those coordinates the complex-time evolution is obtained simply by multiplying the (m, n)-coefficient by $e^{iz(\log m - \log n)}$.

The construction is deliberately elementary: finite unique factorization, unilateral shifts, a diagonal logarithmic dynamics, Gibbs weights, and the KMS condition suffice. In this form the finite Euler factors appear not as a formal analogy but as the literal partition function of a concrete Toeplitz dynamical system.

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